

SUWETHA B

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PROFESSIONAL SUMMARY

B.Tech Information Technology student (CGPA 8.4) who builds AI-integrated products end-to-end — from GraphSAGE-based ML systems to production-style SaaS tools — backed by full-stack MERN development and DevOps/SRE experience from internships at IGenuine Technologies and vSphere Technologies. Frequent hackathon team lead with a 2nd Runner-up finish at JCT SDG Hackathon 2026 and a Top 100 placement (of 800+ teams) in StartupTN & HCL Foundation's AI For Social Impact Challenge. Seeking a Software Engineering internship to ship real product features fast.

TECHNICAL SKILLS

- Languages:** C, C++, Java, JavaScript, Python
- Web Technologies:** HTML, CSS, React.js, TypeScript
- Backend:** Node.js, Express.js
- AI/ML:** OpenAI API (GPT-4o), Groq LLM, GraphSAGE (GNN), scikit-learn, NLP Text Classification
- Databases:** MongoDB, SQL, PostgreSQL
- DevOps & Cloud:** Docker, Kubernetes, Git, GitHub, AWS, CI/CD
- Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Computer Networks, Cloud Computing, System Design
- API & Testing:** REST APIs, Postman, Jest, Pytest

INTERNSHIP EXPERIENCE

- DevOps & SRE Intern — vSphere Technologies** June 2026 – July 2026
- Provisioned and managed virtual machines using VMware, accessing remote systems via MobaXterm to support infrastructure operations.
 - Deployed and managed containerized applications with Docker and Kubernetes across 5+ deployment environments, improving environment consistency across builds.
 - Automated build and deployment workflows using GitLab CI/CD pipelines, cutting manual release steps by roughly 30%.
- MERN Stack Developer Intern — IGenuine Technologies** December 2025
- Built and shipped full-stack features end-to-end across the MERN stack, from schema design through UI integration.
 - Designed and integrated REST APIs connecting frontend and backend systems, coordinating with the existing codebase and team conventions.
 - Used Git for version control in a collaborative development workflow, including code review and branch-based feature work.

PROJECTS

- NoteMind — AI-Powered Research Assistant with RAG-Based Document Chat** [LINK](#) 2026
- Tech Stack: React, TypeScript, Node.js/Express, MongoDB Atlas Vector Search, Groq API (Llama3-70B)
- Built a full-stack RAG system that ingests PDFs/articles through an async pipeline, chunking, embedding (all-MiniLM-L6-v2), and auto-generating summaries, key points, and categories via a single structured LLM call.
 - Implemented semantic cross-document chat with MongoDB Atlas Vector Search (in-memory cosine fallback), streaming responses to the client over SSE with source citations.
 - Designed JWT auth with rotated opaque refresh tokens and an Axios auto-refresh interceptor, keeping sessions alive without exposing long-lived tokens client-side.
 - Shipped a fully typed (TypeScript both ends) UI with folder organization, key-point diffing across document versions, persisted highlights, and one-click Markdown/PDF export.
- Agent Spend Policy Guard — Pre-Settlement Policy Enforcement for Autonomous AI Agent Payments** [LINK](#) 2026
- Tech Stack: FastAPI, Pydantic, Python, x402 Protocol, Vanilla JS Dashboard
- Built a policy enforcement middleware that validates autonomous AI agent transactions against configurable rules before settlement on the x402 payment protocol, preventing unauthorized spend without breaking agent autonomy.

- Implemented a five-rule policy engine — per-transaction spend caps, merchant allowlists, frequency caps, rolling budget windows, and a heuristic risk score derived from amount, category, and vendor patterns.
- Designed a deny-with-reason response contract so rejected agents receive a machine-readable cause instead of an opaque failure, enabling retry or escalation logic. [Add: evaluation latency, e.g. sub-Xms per decision]
- Exposed the engine through a FastAPI service and built a live dashboard with a transaction simulator to stream evaluation results and audit policy decisions in real time.

WildFirePredict OS — Multi-Agent Wildfire Prediction Platform (Ongoing)

2026

Tech Stack: Python, NASA FIRMS API, Cellular Automata, A* Pathfinding, Groq LLM, Flask, Leaflet.js

- Built a multi-agent wildfire prediction platform that models fire spread using cellular automata on real-time NASA FIRMS satellite fire-detection data.
- Integrated A* pathfinding for safe route planning and a Groq-powered LLM agent for natural-language risk summaries — won 2nd Runner-up at JCT SDG Hackathon 2026 as team lead.
- Structured the system as a five-phase autonomous agent pipeline — data ingestion, spread modelling, risk scoring, route planning, and LLM summarization — enabling end-to-end automation from satellite data to actionable insight.
- Built an interactive Leaflet.js dashboard to visualize live fire-spread predictions and safe evacuation routes on a map interface.

RumorGuard — GNN-Based Misinformation Propagation Predictor for Disaster Networks | [LINK](#)

2026

Tech Stack: GraphSAGE (GNN), NLP Text Classification, Python, Cytoscape.js

- Implemented an ML system combining GraphSAGE (GNN) and NLP-based text classification to predict how disaster-related misinformation spreads across mesh networks. [Add: model accuracy/F1 score]
- Generated a combined risk score from model outputs to help responders prioritize which rumors to fact-check first.
- Built a hybrid NLP classification layer combining a transformer-based text classifier with LLM-assisted labeling for higher-precision misinformation detection.
- Served the trained model through a Flask API and built an interactive Cytoscape.js dashboard to visualize propagation paths and risk scores in real time.

ACHIEVEMENTS

- 2nd Runner-up, JCT SDG Hackathon 2026.
- Top 100 Finalist - AI For Social Impact Challenge, conducted by StartupTN and HCL Foundation at Anna University.
- Finalist - Agentica 2.0 (60-hour hackathon), conducted by IIIT Sri City, Andhra Pradesh.
- Finalist - Selfie Hackathon, Sri Eshwar College of Engineering.
- Finalist - Coimbatore Innovation Hackathon, Rathinam Technical Campus 2026.

COMPETITIVE PROGRAMMING

LeetCode: Solved 200+ problems across Data Structures & Algorithms · 1 badge | [LINK](#)

SkillRack: Solved 900+ problems · 10+ certifications · 250+ Bronze badges · Global Rank 28,582 | [LINK](#)

CodeChef: Solved 110+ problems · 1 badge | [LINK](#)

CERTIFICATIONS

Mastering Data Structures and Algorithms Using C and C++, Udemy — 2026

The Complete Java Programmer: From Scratch to Advanced, Udemy — 2025

DBMS: Top Interview Questions, CodeChef — 2025

C, C++: Complete Spoken Tutorial Training, IIT Bombay — 2024

In Progress: AWS Certified Cloud Practitioner (CLF-C02), AWS Training & Certification — 2026

EDUCATION

Sri Eshwar College of Engineering — B.Tech Information Technology, CGPA: 8.4 (up to 4th semester) | 2024 – 2028

AVB Matriculation Higher Secondary School — HSC: 89.0% | 2023 – 2024

AVB Matriculation Higher Secondary School — SSLC: 91.0% | 2021 – 2022